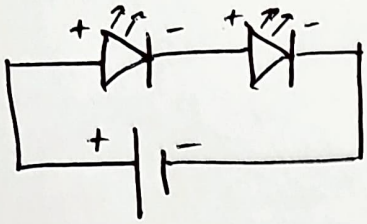
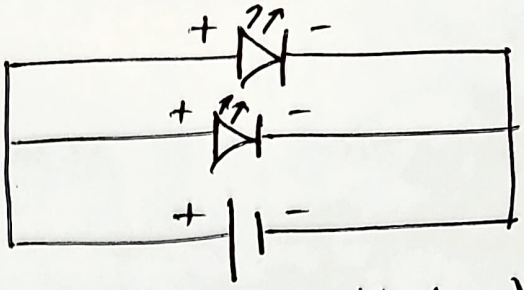
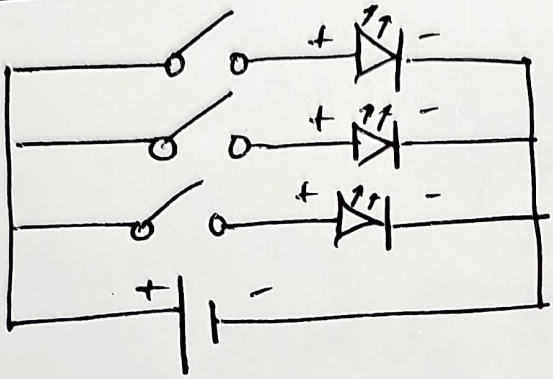
(LED in series)



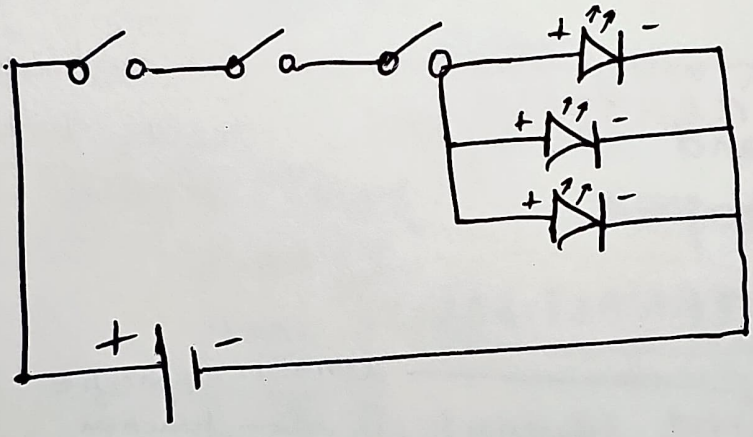
(LED in parallel)



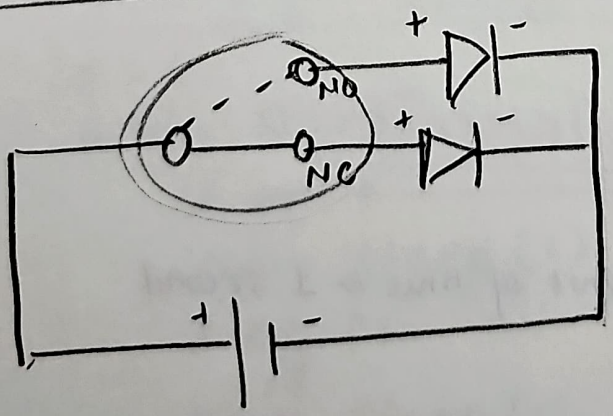
(3 LEDs w/ independent switches)



(3 LEDs that depend on all 3 switches)



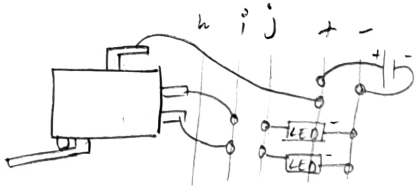
LIMIT SWITCH



NO: Normally open (triggers only when switch is clicked)

NC: Normally closed

LIMIT SWITCH (ON breadboard)



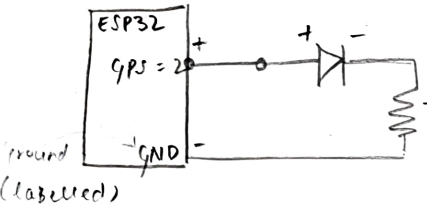
Bread board common connections

abcde, Fghij (rows)

+ column

- column (usually half)

ESP32 diagram in circuit



→ Use resistor calculator online and calculate band resistance

(*) Gold and silver @ end coz they denote tolerance.

connect in a way that fg of LHS and cde of RHS are unoccupied.

LIGHTING INTERNAL LED

```
from machine import Pin
l = Pin(2, Pin.OUT)
l.on()
```

library

specific function

telling device to access LED no. (2).

BLINKING | DELAY of internal LED

```
import time
time.sleep(1)
l.off()
```

Unit of time ⇒ 1 second.

Manual looping ⇒ keep repeating on, sleep, off (blinking)

LOOPING of internal LED (with while)

```
from machine import Pin
import time
l = Pin(2, Pin.OUT)
n = 4
while (n>3):
    l.on()
    time.sleep(1)
    l.off()
    time.sleep(1)
```

} the point of this is to provide a condition that is ALWAYS true.

∴ the LED will keep ~~bringin~~ blinking. (runs forever)

alternative

```
while True:
    l.on()
    time.sleep(1)
    l.off()
    time.sleep(1)
```

case-sensitive.

blinking for specific amount of times

```
n = 6
while (n>3):
    l.on()
    time.sleep(1)
    l.off()
    time.sleep(1)
    n -= 1
```

OUTPUT

6 → 5 → 4

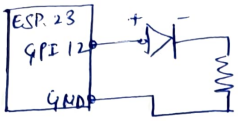
blinks (3) times.

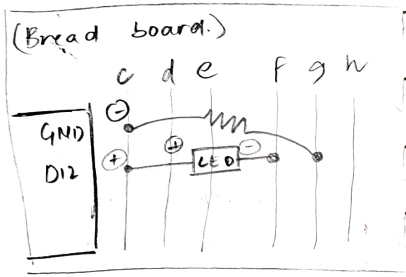
ESP 23 w/ EXTERNAL LED (looping)

```
from machine import Pin
import time
l = Pin(12, Pin.OUT)
while True:
    l.on()
    time.sleep(0.25)
    l.off()
    time.sleep(0.25)
    l.on()
    time.sleep(0.1)
    l.off()
    time.sleep(0.1)
```

pin no. on board

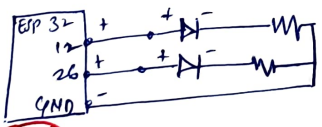
connect →





(*) (↑) resistance → (↓) intensity of LED (proven by changing resistors)

$$\text{resistance} \propto \frac{1}{\text{intensity}}$$



Not gonna work coz voltage ain't enough.

(*) Avoid pin 2 for external coz internal led.

Q Only 5 GND points in total, 3 RHS + 2 LHS, how do we add more LEDs?

⇒ NO, use jumper wire to connect to last section of bread board (⊖ the column)

↓

∴ You can add alot.

ESP 23 w/ MULTIPLE EXTERNAL LEDs

```
from machine import Pin
import time
l = Pin(12, Pin.OUT)
x = Pin(14, Pin.OUT)
y = Pin(27, Pin.OUT)
z = Pin(26, Pin.OUT)
while True:
    l.on()
    time.sleep(1)
    l.off()
    time.sleep(1)
    x.on()
    time.sleep(0.25)
    x.off()
    time.sleep(0.25)
    y.on()
    time.sleep(0.5)
    y.off()
    time.sleep(0.5)
    z.on()
    time.sleep(0.25)
    z.off()
    time.sleep(0.25)
```